Prepared by the Data Science team of Continuum Insights.

04/04/2026

Circular Economy **decision engine**

Turning Returns into Resources:
Leveraging Machine Learning for a Sustainable Product Lifecycle

Contents

Continuum Insights

E-commerce Returns

To transform electronic retail returns from a 'Value Leak' into a 'Recovery Stream' by deploying a Decision Engine that predicts secondary market floor prices.

e-Commerce Returns:
A Billion-Dollar Leak

In 2024 alone, retailers lost over $100 billion to return fraud and inefficiency.

Digital vs In store Returns

Online sales are returned at a rate of 24.5%, compared to only 8.7% of brick-and-mortar purchases.

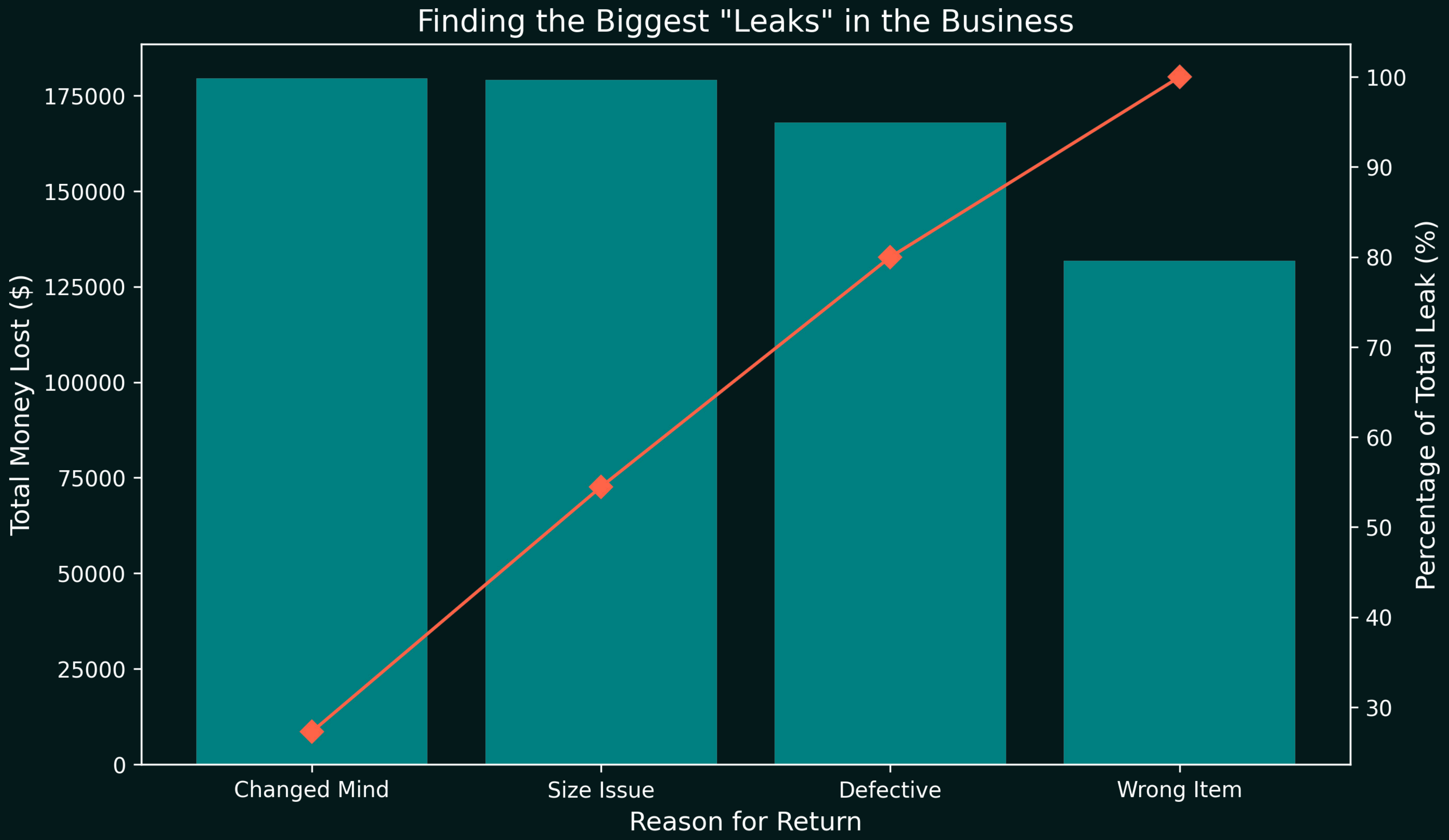
Profit Leaks

A single return costs retailers between 20% to 65% of the item's original value.

The chart shows an example of the price retailers pay when it comes to electronic returns for a laptop.

Expense Category	Cost to Retailer	% of Original Value
Logistics Fee (Shipping + Labor)	$20.00	20%
Open-Box Markdown	$30.00	30%
Packaging/ Refurbishing Materials	$5.00	5%
Customer Acquisition Cost (Wasted Ad Spend)	$10.00	10%
Total Value Leaked	$65.00	65%

Visualizing the "Value Leak" in Experimental Data



Our Objective

Our objective at Continuum Insights is simple: Plug the leak. We have developed a Decision Engine that moves us from 'manual uncertainty' to 'automated precision'.

The Client Interface

The Onboarding Process

- **Connecting an API to your existing Warehouse Management System (WMS) & ERP.** This allows our engine to see returns the moment they are scanned at your loading dock.

- **Data Ingestion:** Providing us with historical "Return Reason" data & current "Hardware Spec" database. This information allows our engine to "learn" the specific lifecycle of your products.

- **Defining "Business Rules":** You can set your own "Risk Appetite". For example, if a client would like any device with residual value under $50 to automatically be recycled, while anything over $150 be sent for manual inspection.

Operational Expections

"The Day-to-Day"

Once the engine is live, retailer's warehouse and logistics teams will see immediate changes in their workflow.

- **The Instant Decision**
  - Engine provides an instant instruction:
    - **Refurbish**
    - **Harvest for Parts**
    - **Recycle**
- **Reduced 'Limbo' Time.**
- **Dynamic Pricing.**
  - Engine suggest the Secondary Market Floor Price in real-time.

Long-Term Value & Reporting

"The ROI"

Profit Recovery

A clear report showing how much of the "20%-65% Value Leak" was plugged, showing the Found Money from reselling items that would've been scrapped.

ESG & Carbon Credits

Precise data on diverted e-waste. For every device refurbished instead of manufactured new, the client can claim a specific reduction in their Scope 3 CO_2 emissions.

Predictive Insights for R&D

The client's product team will expect "Feedback Loops". If our engine shows a specific hardware causing 40% of all returns, they will use the data to improve the design of next year's model.

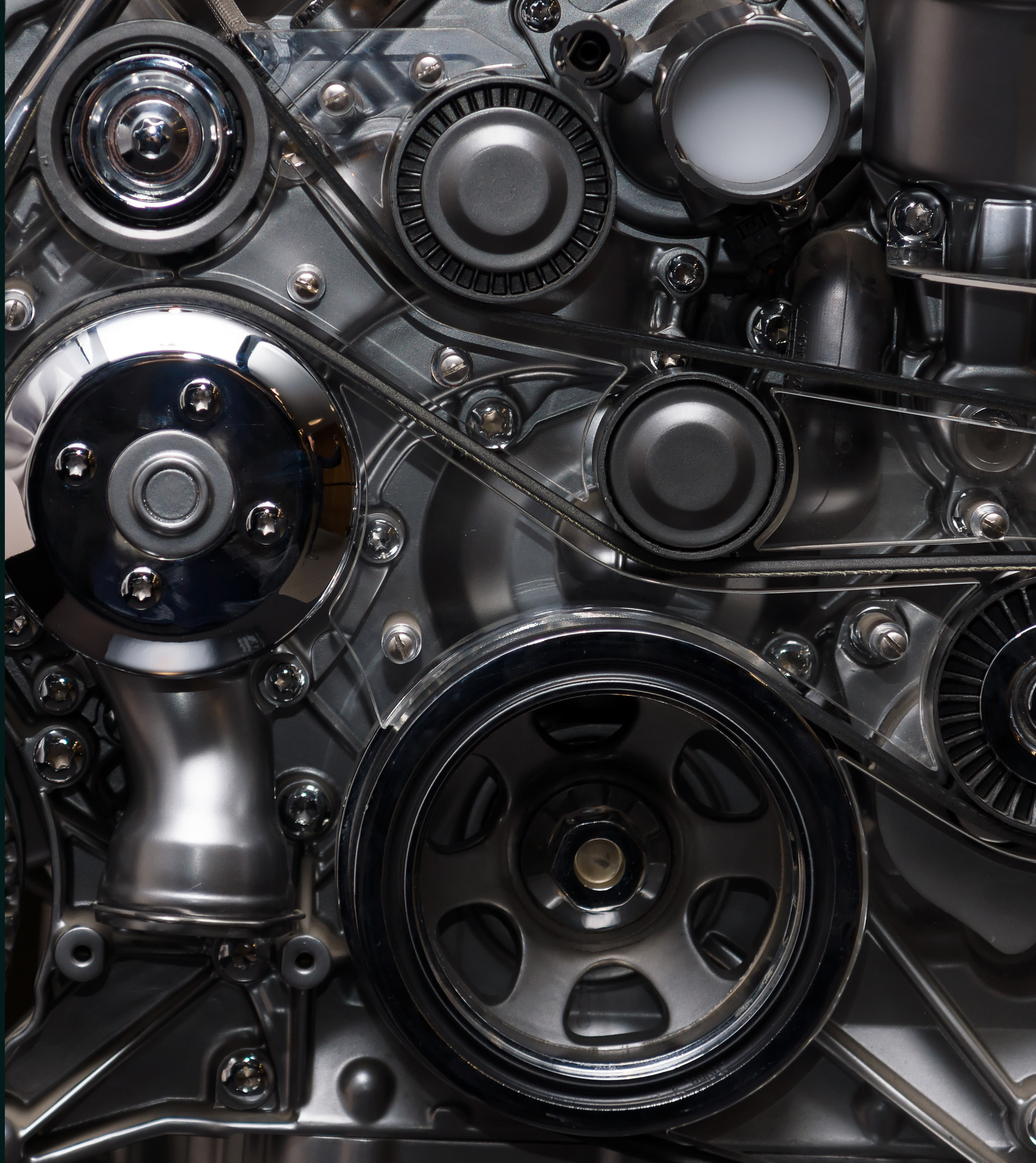
Inside the Engine

The Brains behind the Engine

Our engine doesn't just look at a return; it performs a two-step audit.

Stage 1-Filters for the likelihood of a return (Classification).

Stage 2-Calculates the exact recovery value of that device (Regression).



The Data Ecosystem:

These are high-impact pillars which directly relate to the decision engine.

Retail Returns & Logistics

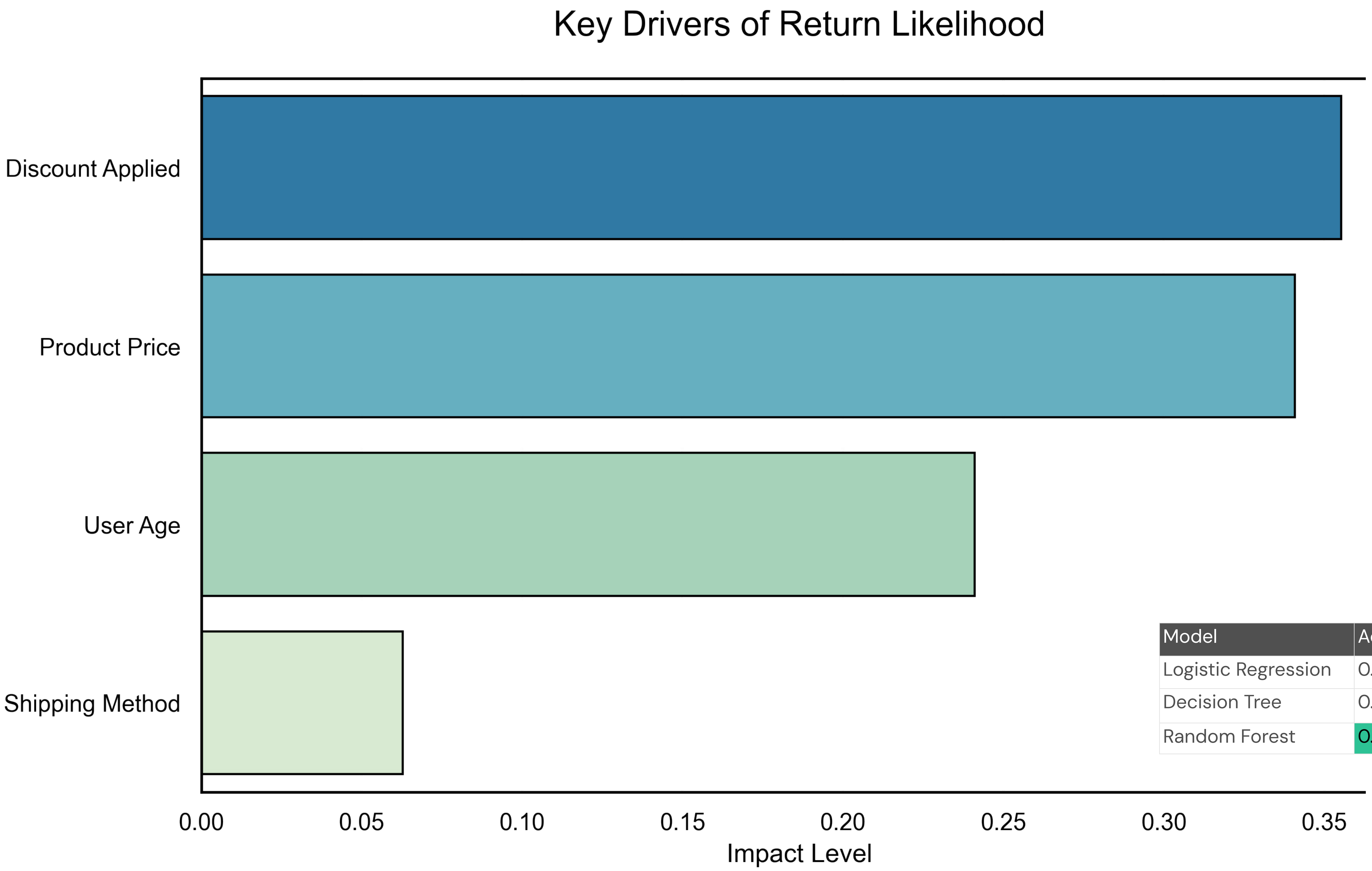
Integrated real-time data from retail return centers.** This allows us to know what devices are coming back.

Hardware Specifications

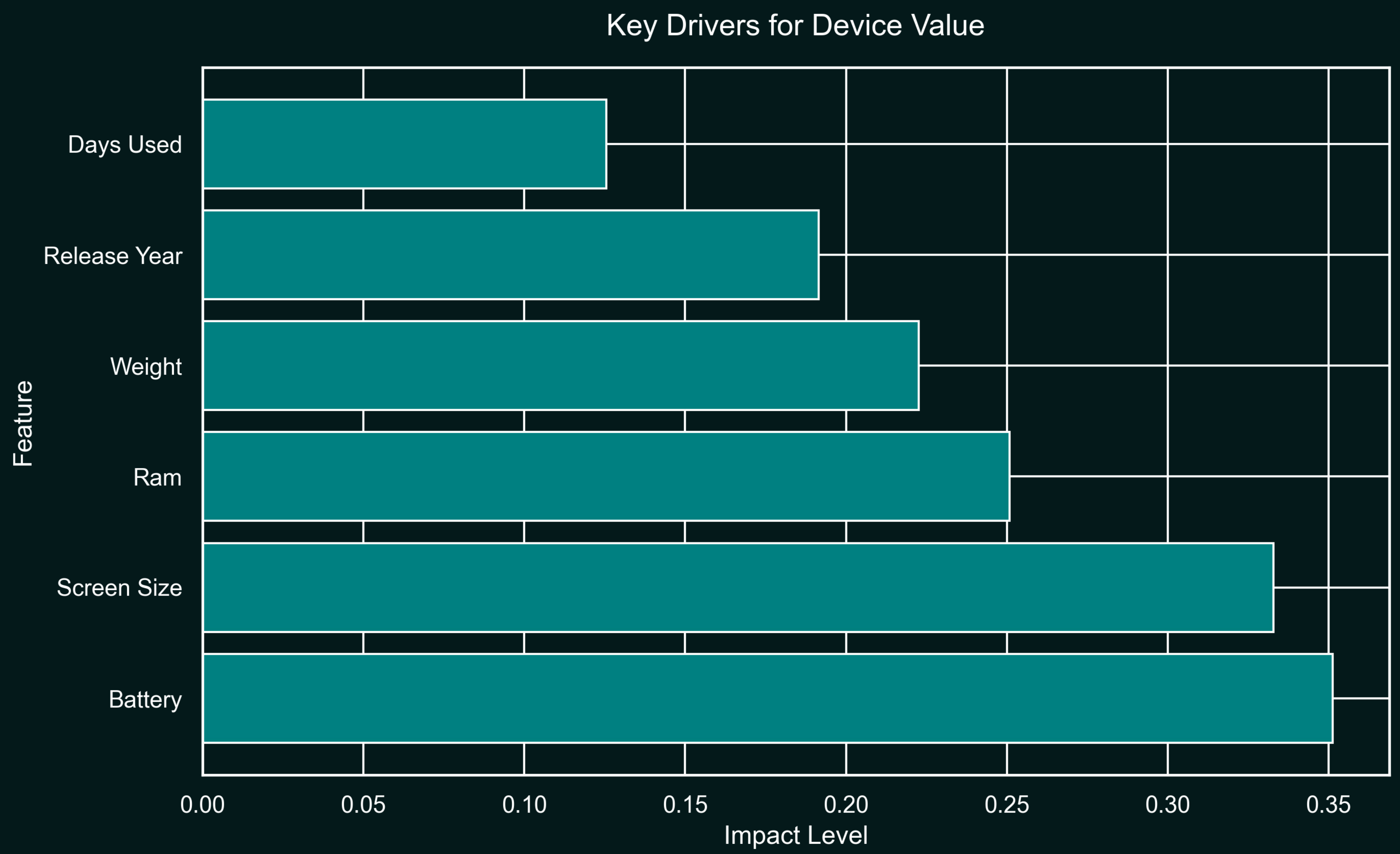
Analyzing component-level data such as Battery Health, Storage, etc. to establish a baseline score.

Predictive Valuation Modeling

Utilizing a dual-stage ML framework to calculate Value-at-Risk and predict the secondary market prices within a specified range.



Model	Accuracy	Recall	F1-Score
Logistic Regression	0.530120	0.492308	0.353591
Decision Tree	0.477912	0.661538	0.398148
Random Forest	0.722892	0.107692	0.168675



Model	MAE
Linear Regression	0.318995
Decision Tree	0.309293
Random Forest	0.270348

Conclusion & Next Steps

In summary, we are turning E-waste into ESG Gold. We are plugging the Billion-Dollar Leak, one 27-cent prediction at a time. My next steps involve a pilot program to integrate this engine directly into live warehouse management systems.

Thank you!